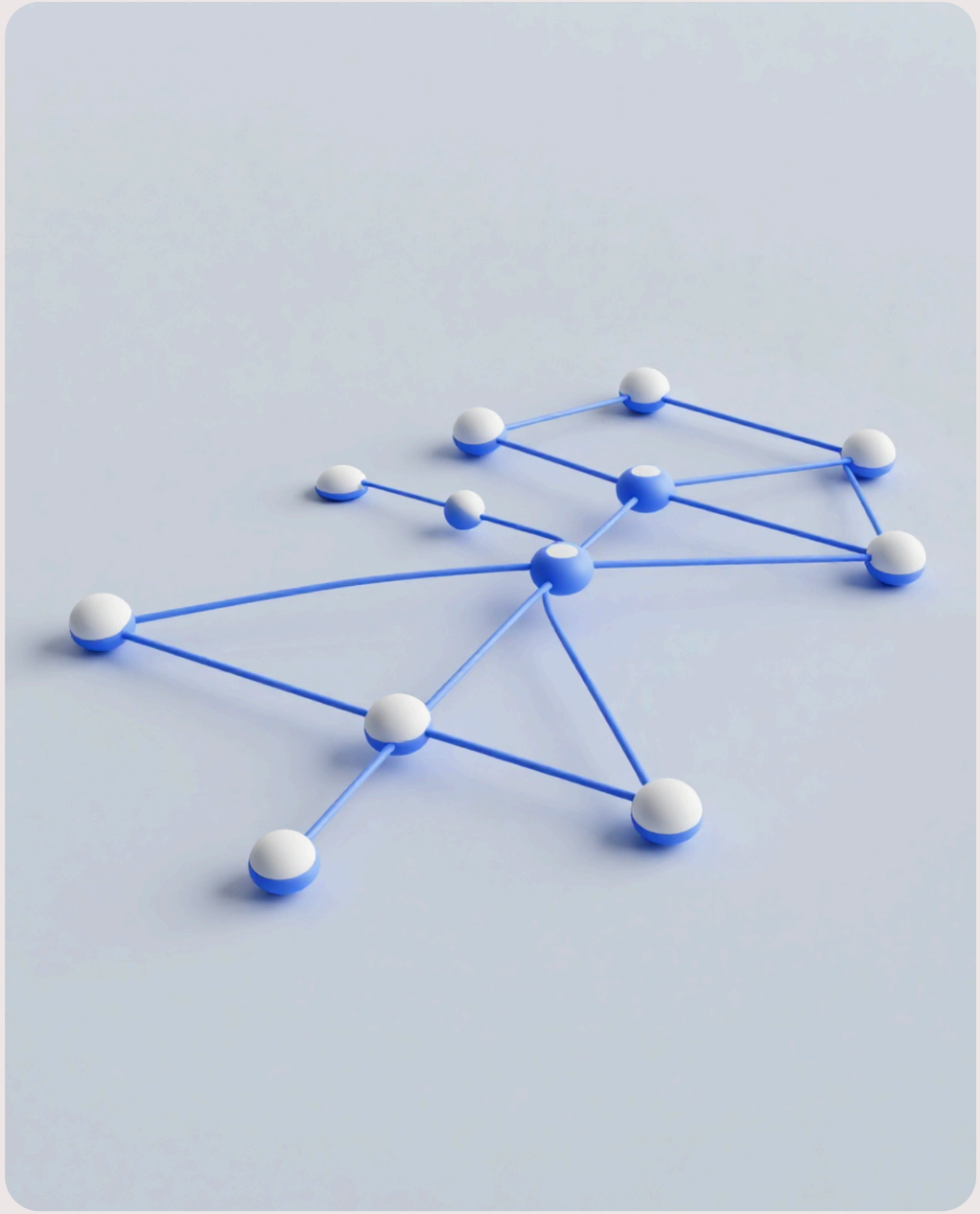
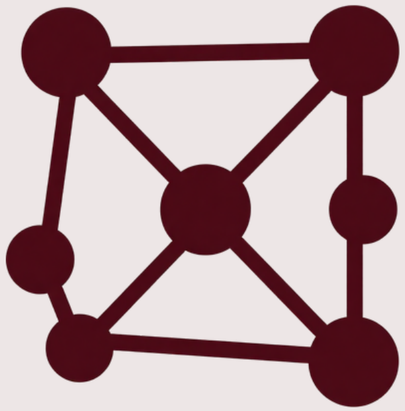
INTRODUCTION

Structured Progress Management in Online Learning

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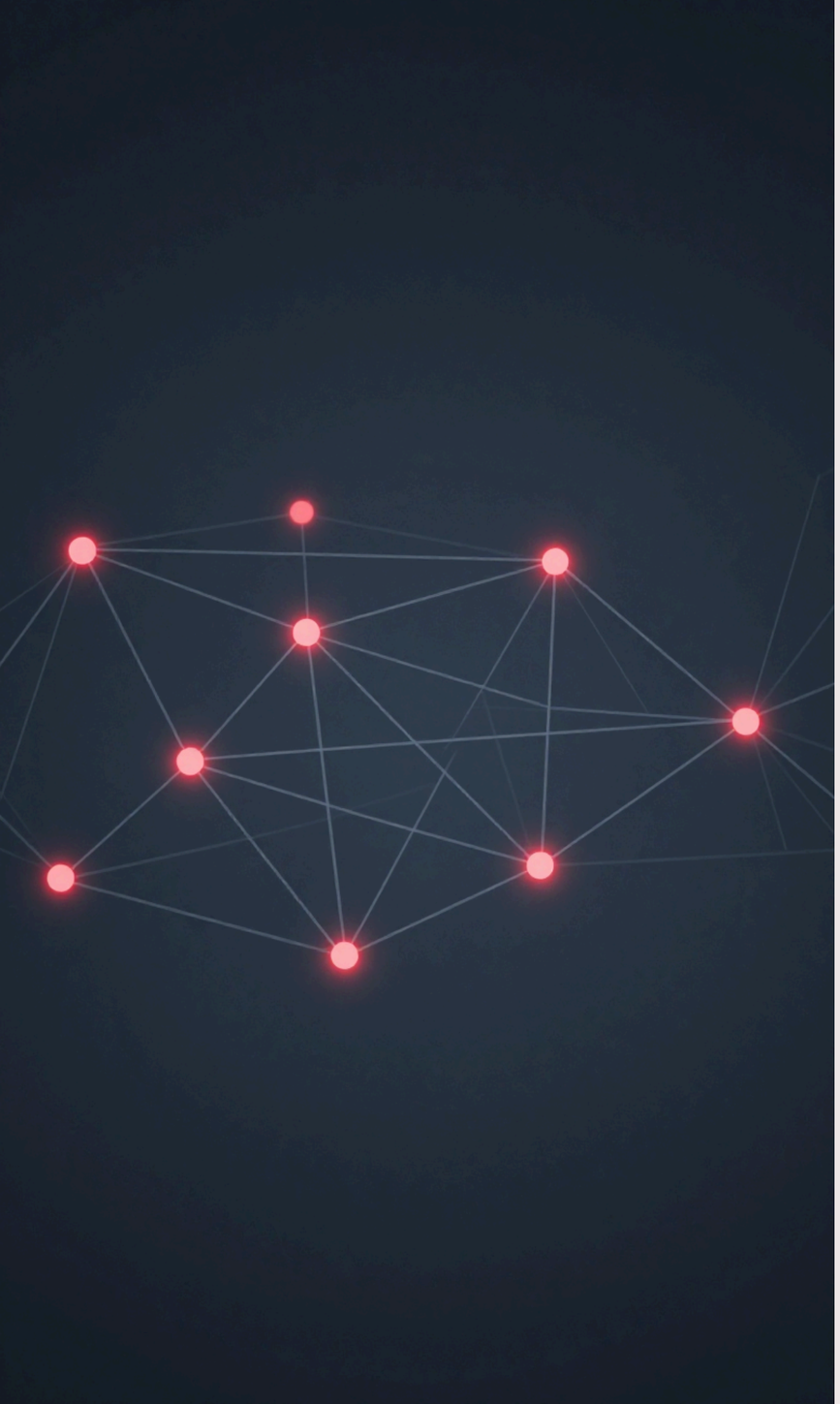
Problem Statement

Core Problem

• Learner progress data is fragmented across multiple platforms – LMS, spreadsheets, and manual records – with no single source of truth.
• Students lack a unified view of completed lessons, assessment scores, deadlines, and overall academic standing.
• Faculty cannot efficiently monitor engagement, identify inactive learners, or flag students at risk of falling behind.
• Manual tracking processes are error-prone, time-consuming, and difficult to verify at scale.

The Central Gap

• Available data exists but is siloed: no mechanism connects enrollment, lesson completion, assessment results, and progress into one actionable view.
• The gap between raw data and actionable information prevents timely academic intervention.
• Institutions lack the infrastructure to generate reliable, real-time progress reports without significant manual effort.
• Without structured data management, early warning signals for struggling learners go undetected until it is too late.



Problem Analysis: Impact

Cause & Effect: Students, Faculty, and Institutions

Students face delayed feedback and unclear progress paths, reducing engagement and self-regulation. Faculty experience increased administrative effort due to manual tracking, limiting timely intervention. Institutions face data-quality risks and struggle to identify at-risk learners early. Evidence-based findings consistently show that fragmented data systems delay corrective action and weaken institutional accountability.